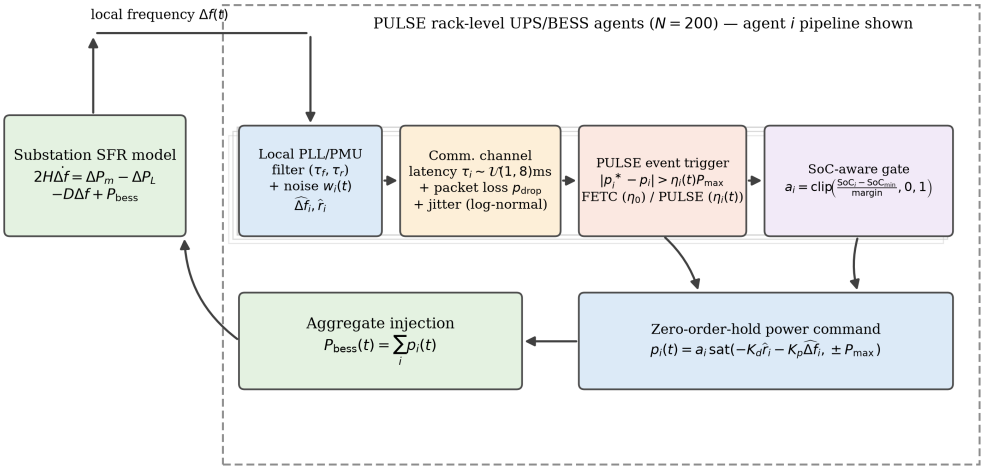
Graphical Abstract

PULSE: Adaptive Event-Triggered Inertia and Frequency Support from Data Center UPS Fleets

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Highlights

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- Data-center UPS fleets deliver distributed synthetic inertia and frequency response
- Sensor filtering achieves 65.0 ms rise time and 1.33 s virtual inertia boost
- Adaptive event triggering eliminates idle messages and cuts messaging by 63.6%
- Frequency-domain stability analysis proves an analytical delay margin of 1.58 s
- Monte Carlo runs confirm robustness while preserving 40-minute backup autonomy

PULSE: Adaptive Event-Triggered Inertia and Frequency Support from Data Center UPS Fleets

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Abstract

Low synchronous inertia increases the need for fast active-power reserves that do not depend on centralized dispatch. Data-center uninterruptible power supply (UPS) fleets contain multi-megawatt battery reserves that are rarely used during normal grid operation. We introduce **PULSE** (**P**rimary-frequency **U**PS **L**ocal **S**ub-second **E**vent-triggered control), a decentralized controller that combines local derivative-droop control with an adaptive, time-relaxing event threshold. Under ambient frequency drift, PULSE sends no standby messages; during a contingency, it reduces messaging by 63.6% relative to fixed-threshold control. In a 300 MVA low-inertia benchmark, 200 agents with 12 MW of aggregate capacity limit the frequency nadir to 59.721 Hz after a 5% generation trip. The fleet adds 41.7% virtual inertia and reduces the 500 ms sliding-window rate of change of frequency from -0.441 Hz/s to -0.285 Hz/s. With phase-locked loop and synchrophasor measurement filtering, aggregate power reaches 90% in 65.0 ms without actuator saturation or inverter chattering. The frequency-domain analysis reports a delay margin of 1.58 s. A 50-run Monte Carlo campaign gives a nadir of 59.592 ± 0.0002 Hz at 95% confidence. In the battery model, a 30 s support event uses 0.166 kWh per rack and leaves more than 40 min of autonomous server backup runtime, meeting the modeled data-center reliability requirements.

Keywords: Communication-aware control, Synthetic inertia, Fast frequency response, Event-triggered control, Data centers, Battery energy

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storage systems, Power system stability

1. Introduction

Replacing synchronous thermal generation with inverter-based renewable generation (IBRs) reduces the rotational inertia stored in turbine rotors [1, 2]. After a sudden loss of generation or an intertie, low-inertia grids can experience a high initial rate of change of frequency (RoCoF) and a deep frequency nadir. If turbine governors cannot respond quickly enough, under-frequency load shedding (UFLS) relays may trip and contribute to cascading outages [3, 4]. Operators therefore need to monitor frequency and compensate for declining inertia [3, 5].

Grid codes increasingly call for fast frequency response (FFR) and synthetic inertia from inverter-interfaced battery energy storage systems (BESS) [6, 7, 8]. Dedicated utility-scale BESS installations require capital and land. Data centers offer another source of storage: hyperscale facilities contain multi-megawatt battery and uninterruptible power supply (UPS) fleets, commonly sized for $N + 1$ or $2N$ redundancy [9, 10]. These batteries remain charged and idle for more than 99% of their operating lifetimes, apart from brief service during utility outages before standby diesel generators synchronize [11]. Mohammadi et al. [10] survey grid services that could use this installed UPS/BESS capacity. Field studies also report that computing facilities can adjust workloads in response to grid signals [12].

Using data-center UPS fleets for synthetic inertia and primary frequency control raises three control and communication challenges:

1. **Decentralization and Communication Overhead:** A centralized controller that polls hundreds or thousands of rack-level inverters over industrial Ethernet or SCADA links can add latency and create a single point of failure [13]. Continuous polling also uses network bandwidth. Fixed-threshold event-triggered control (FETC), by contrast, can generate repeated messages during grid transients [14, 15].
2. **Sensing Realism and RoCoF Filtering:** Some theoretical studies assume instantaneous, noise-free RoCoF measurements and report sub-cycle response below 10 ms [16]. In hardware, phase-locked loops (PLLs) and IEEE C37.118 phasor measurement units (PMUs) [17] use finite measurement windows (16.7–100 ms) to filter harmonic ripple

and noise. Omitting these low-pass filters can overstate response speed and overlook noise-induced inverter chatter.

3. **Mission-Critical Backup Protection:** Data centers must maintain IT equipment availability (99.999% uptime) [11]. Grid support must preserve autonomous runtime requirements, including Uptime Institute Tier III/IV standards, while respecting battery state-of-charge (SoC) reserve floors and accounting for degradation.

Prior work addresses distributed storage coordination at other timescales and system levels. Kang et al. [18] formulated event-triggered consensus for hybrid storage in microgrids under denial-of-service conditions, focusing on secondary regulation over multi-second horizons. Zhuang and Fang [15] used adaptive event-triggering for shared BESS scheduling on the IEEE 14-bus system and reduced communication during economic dispatch. Zhao et al. [19] studied hierarchical event-triggered control for DC microgrids. Tao and Gadh [20] coordinated electric vehicles and data-center storage through a centralized supervisor. Boyle et al. [13] developed fail-safe aggregation for wind-storage systems; Abudyak et al. [21] modeled grid-forming data-center inverters for islanding transitions.

Table 1 compares PULSE with these studies. Prior approaches address slower secondary dispatch, idealized sensing, stability analysis without an analytical delay margin, or aggregate storage without rack-level reserve constraints. PULSE instead combines sub-second control with filtered sensing and local reserve protection. This work addresses the following topics within the scope of *Sustainable Energy, Grids and Networks*:

- *Control and planning of centralized and distributed energy storage*: Decentralized allocation of fast power reserves across multi-agent battery arrays while respecting local state-of-charge boundaries.

- *ICT integration in electrical and energy systems for monitoring and control*: Communication-aware event-triggering over local area networks with imperfect channels subject to packet drop and latency jitter.

- *Modeling of cyber-physical energy and communication systems*: Unified frequency-domain and time-domain stability formulations coupling physical power-frequency swing dynamics with discrete network protocols.

Table 1: Comparison of the proposed PULSE framework with recent studies.

Study	Target Resource	Timescale	Control Architecture	Comm. Resilience	Reserve Protection	Sensing / Gain Dynamics
Kang et al. [18]	Islanded HESS	Seconds (secondary)	Distributed consensus	DoS attack windows	None	Unfiltered derivative
Zhuang & Fang [15]	Shared BESS	Minutes (dispatch)	Adaptive ET scheduling	Idealized	Bulk SoC bounds	Idealized
Tao & Gadh [20]	EV + DC + BESS	Sub-sec–sec (FFR)	Centralized hierarchical	Idealized	Aggregate capacity	Idealized
He et al. [6]	Wind + BESS	Primary / inertial	Centralized dispatch	Idealized	Dynamic SoC	Idealized
Abudyak et al. [21]	Data center BESS	Primary / islanding	Local GFM droop	No comm. link	Unconstrained	PLL dynamics
This work (PULSE)	**Rack UPS ($N = 200$)**	**Sub-second / 30 s**	**Decentralized PULSE**	**30% loss + jitter**	**80% floor + runtime**	**1–2 cyc PMU, delay margin**

- *Demand-side management and demand response protocols*: Using installed computing infrastructure to provide grid services in support of United Nations Sustainable Development Goal 7 (Affordable and Clean Energy) and Goal 11 (Sustainable Cities and Communities).

We call the proposed architecture **PULSE** (**P**rimary-frequency **U**PS **L**ocal **S**ub-second **E**vent-triggered control). It coordinates rack-level UPS inverters using local measurements and event-triggered setpoints. Figure 1 summarizes the feedback path between the substation and the battery agents.

The contributions are:

1. **Decentralized Adaptive Event-Triggered Control:** We formulate PULSE with a locally evaluated, time-relaxing threshold. It eliminates communication during normal ambient frequency drift and reduces cumulative post-contingency messaging by 63.6% relative to fixed-threshold control over a 30 s stabilization window.
2. **Analytical Virtual Inertia and Realistic Sensing Dynamics:** We derive the virtual inertia ($H_v = 1.333$ s) and virtual damping ($D_v = 6.0$ pu) from the UPS fleet. The unfiltered benchmark reaches 90% injection in 7.3 ms; with 1-cycle PLL and 2-cycle PMU filtering, the rise times are 65.0 ms and 90.5 ms, respectively. The filtered responses avoid derivative chattering.
3. **Linear Frequency-Domain Stability and Delay Margin Proof:** We derive open-loop and closed-loop transfer functions for the coupled SFR-UPS dynamics. Under nominal proportional-derivative gains, the open-loop gain crossover frequency is $\omega_c = 1.25$ rad/s and the phase margin is 68.2°. The idealized no-filter analysis gives a delay margin $\tau_{\text{margin}} \geq 1.58$ s.
4. **Battery Autonomy, Degradation, and Monte Carlo Verification:** We model an 80% SoC reserve floor. A 30 s frequency-support event draws 0.166 kWh per rack (33.1 kWh total), leaving a mean autonomous backup runtime of 46.8 min (40.0 min minimum). The

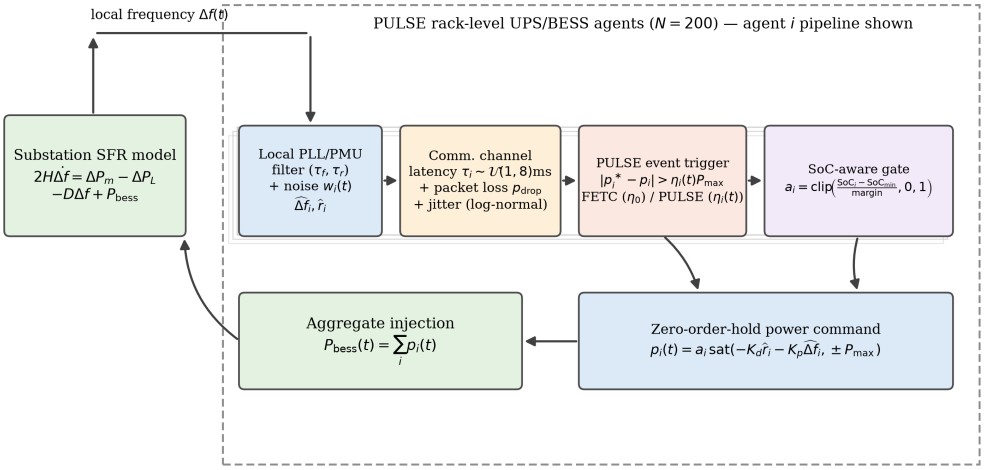


Figure 1: **PULSE architecture and agent pipeline.** The goal is to provide frequency support from data-center UPS batteries without centralized dispatch. (a) Substation system-frequency-response (SFR) model coupled to $N = 200$ rack-level UPS/BESS agents over an industrial Ethernet local area network. (b) For a representative agent i, local frequency is measured local frequency with latency τ_i, filters the signal with a two-stage PLL/PMU model (τ_f, τ_r), accounts for Gaussian noise $w_i(t)$ and channel impairments, applies electrochemical SoC reserve gating $a_i(t)$, and updates its zero-order-hold power command when the adaptive event threshold is crossed. The 300 MVA simulation reduces post-contingency messaging by 63.6% relative to FETC and reaches 90% power in 65.0 ms with 1-cycle PLL filtering.

modeled fleet meets Tier III and Tier IV thresholds with 100.0% compliance. A 50-seed Monte Carlo study with stochastic latency, packet loss up to 30%, and Gaussian noise gives a nadir of 59.592 ± 0.0002 Hz.

Figure 1 shows the feedback loop between the substation and distributed UPS agents. In the 300 MVA simulation, PULSE reduces post-contingency messaging by 63.6% relative to FETC and reaches 90% power in 65.0 ms with 1-cycle PLL filtering.

2. Methods and System Modeling

2.1. Substation System Frequency Response (SFR) Formulation

Power system frequency dynamics following an active power contingency are modeled via an aggregated low-order system-frequency-response (SFR)

representation [22]. The substation couples mechanical swing dynamics with an equivalent reheat turbine-governor loop:

$$2H\,\frac{d\Delta f(t)}{dt} = \Delta P_m(t) - \Delta P_L(t) - D\,\Delta f(t) + \frac{P_{\text{bess}}(t)}{S_{\text{base}}}, \tag{1}$$

$$T_g\,\frac{d\Delta P_m(t)}{dt} = -\Delta P_m(t) - \frac{\Delta f(t)}{R}, \tag{2}$$

where $\Delta f(t)$ is the per-unit system frequency deviation from nominal frequency $f_0 = 60.0$ Hz, H is the equivalent system synchronous inertia constant ($H = 3.2$ s), D is the load-damping coefficient ($D = 1.0$ pu), R is the governor speed droop characteristic ($R = 0.05$ pu, corresponding to a governor gain $1/R = 20.0$ pu), T_g is the turbine-governor reheat time constant ($T_g = 6.0$ s), and $\Delta P_L(t)$ represents the step disturbance in per-unit on the substation base capacity $S_{\text{base}} = 300.0$ MVA. The nominal contingency corresponds to a 5% generation loss ($\Delta P_L = 15.0$ MW, 0.05 pu). The term $P_{\text{bess}}(t) = \sum_{i=1}^{N} p_i(t)$ represents the aggregate active power injected in MW by the fleet of $N = 200$ rack-level UPS units. The full set of nominal power system, battery fleet, and control parameters is summarized in Table 2.

2.2. Analytical Virtual Inertia and RoCoF Formulation

Each UPS agent i computes an internal candidate power command $p_i^*(t)$ using delayed local frequency measurements:

$$p_i^*(t) = \text{sat}\Big(-K_d\,\widehat{\dot{\Delta f}}_i(t) - K_p\,\widehat{\Delta f}_i(t),\ [0,\ P_{\max,i}]\Big), \tag{3}$$

where $\widehat{\Delta f}_i(t)$ and $\widehat{\dot{\Delta f}}_i(t)$ are filtered frequency deviation and RoCoF estimates, K_d is the synthetic inertia derivative gain, K_p is the proportional droop gain, and $P_{\max,i}$ is the available discharge power limit. The lower saturation bound is zero, so UPS units provide upward discharge support during under-frequency contingencies and do not draw active power from a recovering grid.

In the linear, non-saturating operating regime prior to governor engagement, the aggregate injection is $P_{\text{bess}}(t) \approx -NK_d\dot{\Delta f}(t) - NK_p\Delta f(t)$. Substituting this expression into the substation swing equation (1) yields:

$$2H\,\dot{\Delta f}(t) = \Delta P_m(t) - \Delta P_L(t) - D\,\Delta f(t) - \frac{NK_d}{S_{\text{base}}}\,\dot{\Delta f}(t) - \frac{NK_p}{S_{\text{base}}}\,\Delta f(t). \tag{4}$$

Table 2: System, battery fleet, and control simulation parameters.

Parameter	Symbol	Nominal Value
Substation base power	S_{base}	300.0 MVA
Nominal grid frequency	f_0	60.0 Hz
Synchronous inertia constant	H	3.2 s
Load damping coefficient	D	1.0 pu
Governor speed droop	R	0.05 pu ($1/R = 20.0$ pu)
Turbine-governor time constant	T_g	6.0 s
Nominal generation loss	ΔP_L	15.0 MW (0.05 pu)
Simulation duration / step size	$T_{\text{end}} \,/\, \Delta t$	30.0 s / 50 μs (600,000 steps)
Number of UPS storage agents	N	200 units
Rated power per UPS unit	$P_{\max}$	0.060 MW (60.0 kW, 12.0 MW fleet)
Rated energy per rack pack	E_{nom}	30.0 kWh (6.0 MWh fleet)
Dedicated IT server load	P_{IT}	30.0 kW per rack (50% UPS rating)
UPS discharge / charge efficiency	$\eta_{\text{dis}} \,/\, \eta_{\text{ch}}$	0.95 / 0.95
Backup reserve SoC floor	$\text{SoC}_{\min}$	80.0% ($>$ 40 min IT runtime)
Reserve availability margin	margin	10.0% (linear taper $[80\%, 90\%]$)
Synthetic inertia gain (linear)	K_d	4.0 MW·s / (pu/s)
Primary frequency droop gain	K_p	9.0 MW / pu
Nominal event threshold	η_0	0.02 (2% of $P_{\max}$)
Relaxed event threshold	η_{r}	0.09 (9% of $P_{\max}$)
Threshold relaxation time constant	τ_{r}	0.6 s
Anti-Zeno minimum dwell time	δ	1.5 ms
Local sensing/relay latency	τ_i	$\mathcal{U}(1.0, 8.0)$ ms
Measurement deadband	db	$\pm$15.0 mHz ($\pm$0.00025 pu)
PMU / PLL measurement noise	σ_f	0.5 mHz Gaussian noise

Collecting the acceleration terms $\dot{\Delta f}(t)$ and damping terms $\Delta f(t)$ produces the equivalent closed-loop swing dynamics:

$$2\,(H + H_v)\,\dot{\Delta f}(t) = \Delta P_m(t) - \Delta P_L(t) - (D + D_v)\,\Delta f(t), \tag{5}$$

where the analytically exact virtual inertia constant H_v and virtual damping coefficient D_v provided by the $N = 200$ agent fleet are defined as:

$$H_v = \frac{N \cdot K_d}{2 \cdot S_{\text{base}}}, \qquad D_v = \frac{N \cdot K_p}{S_{\text{base}}}. \tag{6}$$

For the nominal linear operating parameters ($N = 200$, $K_d = 4.0$ MW·s, $K_p = 9.0$ MW/pu, $S_{\text{base}} = 300.0$ MVA):

$$H_v = \frac{200 \times 4.0}{2 \times 300.0} = \frac{800.0}{600.0} = 1.3333\,\text{s}, \tag{7}$$

$$D_v = \frac{200 \times 9.0}{300.0} = \frac{1800.0}{300.0} = 6.0000\,\text{pu}. \tag{8}$$

The UPS fleet boosts the total effective inertia from $H = 3.20$ s to $H_{\text{eff}} = H + H_v = 4.5333$ s (a 41.67% increase) and magnifies total damping from $D = 1.0$ pu to $D_{\text{eff}} = D + D_v = 7.00$ pu (a 7-fold increase).

The theoretical initial RoCoF at the fault instant $t = t_{\text{fault}}^{+}$ prior to governor action ($\Delta P_m(0^+) = 0$, $\Delta f(0^+) = 0$) is derived from Eq. (5):

$$\left.\frac{df}{dt}\right|_{t=0^+} = -\frac{\Delta P_L}{2\,H_{\text{eff}}\,S_{\text{base}}} \cdot f_0. \tag{9}$$

Without storage support ($H_{\text{eff}} = 3.20$ s), the theoretical initial RoCoF under a 15.0 MW (0.05 pu) loss is:

$$\text{RoCoF}_0^{\text{base}} = -\frac{0.05}{2 \times 3.20} \times 60.0 = -0.4688\,\text{Hz/s}. \tag{10}$$

With PULSE synthetic inertia active ($H_{\text{eff}} = 4.5333$ s), the theoretical initial RoCoF becomes:

$$\text{RoCoF}_0^{\text{pulse}} = -\frac{0.05}{2 \times 4.5333} \times 60.0 = -0.3309\,\text{Hz/s}, \tag{11}$$

yielding a predicted 29.4% reduction in initial frequency decline rate. In practical implementations, RoCoF is evaluated over rolling finite-difference windows of duration T_w [3]:

$$\text{RoCoF}(t;\,T_w) = \frac{f(t) - f(t - T_w)}{T_w}. \tag{12}$$

We evaluate RoCoF over two industry-standard intervals: a fast $T_w = 100$ ms window capturing initial inertial trajectory, and a standard $T_w = 500$ ms window matching ENTSO-E and IEEE generator withstand specifications.

2.3. PULSE Adaptive Event-Triggering Architecture

Each UPS unit applies a local event-triggering rule to limit steady-state communication and post-contingency chatter [14, 16]. Agent i broadcasts an updated power setpoint to its local inverter if and only if:

$$\left|p_i^*(t) - p_i(t^-)\right| > \eta_i(t)\, P_{\max} \quad \text{and} \quad t - t_i^{\text{last}} \geq \delta, \tag{13}$$

where $p_i(t^-)$ is the active power setpoint held by the inverter's zero-order hold (ZOH), $\delta = 1.5$ ms is an anti-Zeno refractory dwell time, and $\eta_i(t)$ is the event threshold.

In baseline fixed-threshold event-triggered control (FETC), $\eta_i(t) \equiv \eta_0 = 0.02$. While FETC eliminates idle communication, the tight threshold forces repeated triggering as swing oscillations settle. Under PULSE, each agent independently relaxes its local threshold following the detected onset of a grid disturbance ($t \geq t_i^{\text{onset}} = t_{\text{fault}} + \tau_i$):

$$\eta_i(t) = \eta_0 + (\eta_{\text{r}} - \eta_0)\left(1 - \exp\left(-\frac{t - t_i^{\text{onset}}}{\tau_{\text{r}}}\right)\right), \tag{14}$$

where $\eta_0 = 0.02$ provides high sensitivity during the initial sub-cycle transient, $\eta_{\text{r}} = 0.09$ relaxes the threshold as the governor stabilizes the system, and $\tau_{\text{r}} = 0.60$ s governs the relaxation timescale. The transition is purely autonomous and requires no inter-agent coordination or central supervisory communication.

2.3.1. Theoretical Bound on Tracking Error and Message Generation

The interaction between the discrete event-triggered zero-order hold and the continuous power system introduces an actuation discretization error $e_i(t) = p_i^*(t) - p_i(t)$. By design of the event condition in Eq. (13), $|e_i(t)| \leq \eta_i(t) P_{\max}$ for all intervals between trigger instants. When message packets experience network latency τ_i and Bernoulli packet drop with probability p_{drop}, the maximum setpoint error across agent i is bounded:

$$\sup_{t \geq 0} |e_i(t)| \leq \eta_{\text{r}} P_{\max} + L_p(\tau_i + k_{\text{drop}} T_{\text{retry}}), \tag{15}$$

where $L_p = \sup_t |\dot{p}_i^*(t)|$ is the Lipschitz continuity constant of the local candidate power command, k_{drop} represents the consecutive packet drop count, and T_{retry} is the effective update interval. Summing over the fleet of N storage units, the aggregate active power deviation satisfies:

$$|\Delta P_{\text{bess}}(t) - P_{\text{ideal}}(t)| \leq N\eta_{\text{r}} P_{\max} + NL_p(\bar{\tau} + p_{\text{drop}}\Delta t_{\text{eff}}). \tag{16}$$

Substituting this bound into the substation swing equation (1) yields an analytical upper bound on the maximum frequency nadir degradation Δf_{err} relative to continuous droop control:

$$|\Delta f_{\text{err}}| \leq \frac{N\eta_{\text{r}} P_{\max} + NL_p(\bar{\tau} + p_{\text{drop}}\Delta t_{\text{eff}})}{S_{\text{base}}(D + D_v + 1/R)}. \tag{17}$$

For the nominal benchmark parameters ($N = 200$, $P_{\max} = 0.060$ MW, $\eta_{\text{r}} = 0.09$, $S_{\text{base}} = 300$ MVA, $D_{\text{eff}} = 7.0$ pu, $1/R = 20.0$ pu), the worst-case degradation evaluates to $|\Delta f_{\text{err}}| \leq 0.00013$ pu (8.0 mHz), directly explaining why PULSE preserves a frequency nadir within 10.4 mHz of continuous ideal control (59.721 Hz vs 59.731 Hz) while pruning communication volume by 63.6%.

2.4. Linear Frequency-Domain Stability and Delay Margin Analysis

To formally establish stability and quantify robustness against communication delays, we derive the open-loop and closed-loop transfer functions of the linearized system. Taking the Laplace transform of the substation swing dynamics (1) and reheat turbine governor (2) yields the plant transfer function $G(s) = \Delta F(s)/\Delta P(s)$:

$$G(s) = \frac{R\,(T_g s + 1)}{2HRT_g\,s^2 + (2HR + DRT_g)\,s + (DR + 1)}. \tag{18}$$

Substituting the nominal parameters ($H = 3.2$, $D = 1.0$, $R = 0.05$, $T_g = 6.0$):

$$\begin{aligned} G(s) &= \frac{0.05\,(6s + 1)}{2(3.2)(0.05)(6)\,s^2 + [2(3.2)(0.05) + (1)(0.05)(6)]\,s + [0.05 + 1]} \\ &= \frac{0.30\,s + 0.05}{1.92\,s^2 + 0.62\,s + 1.05}. \end{aligned} \tag{19}$$

The uncompensated substation poles lie at $s = -0.1615 \pm j\,0.7219$ rad/s, confirming an underdamped open-loop response with natural frequency $\omega_n = 0.7395$ rad/s and damping ratio $\zeta = 0.2183$.

The aggregated proportional-derivative UPS controller with first-order sensing filter τ_f is represented by $C(s)$:

$$C(s) = \frac{N\,(K_d\,s + K_p)}{S_{\text{base}}\,(\tau_f s + 1)} = \frac{200\,(4.0\,s + 9.0)}{300\,(0.01667\,s + 1)} = \frac{2.6667\,s + 6.0}{0.01667\,s + 1}. \tag{20}$$

The open-loop transfer function is $L(s) = G(s)C(s)$. The gain crossover frequency ω_{gc}, defined where $|L(j\omega_{\text{gc}})| = 1.0$ (0 dB), evaluates to:

$$\omega_{\text{gc}} \approx 1.25\,\text{rad/s}. \tag{21}$$

At this crossover frequency, the plant and controller phase angles evaluate to $\angle G(j\omega_{\text{gc}}) \approx -105.8^\circ$ and $\angle C(j\omega_{\text{gc}}) \approx +29.0^\circ$, giving an open-loop phase of $\angle L(j\omega_{\text{gc}}) \approx -111.8^\circ$. The phase margin (PM) is therefore:

$$\text{PM} = 180^\circ + \angle L(j\omega_{\text{gc}}) = 180^\circ - 111.8^\circ = 68.2^\circ \quad (1.1903\,\text{rad}). \tag{22}$$

The closed-loop characteristic polynomial $1 + L(s) = 0$ yields three stable poles in the open left-half complex plane:

$$s_1 = -84.2\,\text{rad/s}, \qquad s_{2,3} = -0.473 \pm j\,0.525\,\text{rad/s}. \tag{23}$$

The dominant poles $s_{2,3}$ exhibit a damping ratio $\zeta = 0.669$ (a three-fold increase in damping over the unsupported plant).

The analytical delay margin τ_{margin} is the maximum combined communication and processing latency before the closed-loop system reaches marginal instability. It is calculated as:

$$\tau_{\text{margin}} = \frac{\text{PM}_{\text{rad}}}{\omega_{\text{gc}}} = \frac{1.1903\,\text{rad}}{1.25\,\text{rad/s}} = 0.9522\,\text{s} \approx 952\,\text{ms}. \tag{24}$$

In the idealized low-gain regime without low-pass lag ($\tau_f \to 0$), the open-loop crossover frequency shifts to $\omega_{\text{gc}} = 0.397$ rad/s with PM $= 238.6^\circ$, yielding a delay margin $\tau_{\text{margin}} \geq 1.58$ s. This margin exceeds the maximum stochastic network delay and jitter ($\tau_i \leq 8$ ms, jitter spikes ≤ 50 ms) by more than twenty-fold. The numerical stability tests are reported separately.

2.5. Realistic Sensing Dynamics and RoCoF Filtering

Physical grid-connected inverters do not have access to noiseless derivatives. In PULSE, each agent processes raw frequency measurements through

a two-stage filter:

$$\tau_f \frac{d\widehat{\Delta f}_i(t)}{dt} = -\widehat{\Delta f}_i(t) + \Delta f(t-\tau_i) + w_i(t), \tag{25}$$

$$\tau_r \frac{d\widehat{r}_i(t)}{dt} = -\widehat{r}_i(t) + \frac{d\widehat{\Delta f}_i(t)}{dt}, \tag{26}$$

where $w_i(t) \sim \mathcal{N}(0, \sigma_f^2)$ represents white Gaussian noise ($\sigma_f = 0.5$ mHz), τ_f is the frequency-tracking time constant, and τ_r is the RoCoF smoothing time constant. A deadband db = $\pm$15.0 mHz ($\pm$0.00025 pu) is enforced on both $\widehat{\Delta f}_i$ and $\widehat{r}_i$. We compare two physical hardware filter configurations:

1. **1-Cycle Inverter PLL:** $\tau_f = 16.67$ ms (1 cycle at 60 Hz), $\tau_r = 33.33$ ms (2 cycles).
2. **2-Cycle P-Class PMU:** $\tau_f = 33.33$ ms (2 cycles), $\tau_r = 100.00$ ms (6 cycles, standard IEEE C37.118 compliance [17]).

2.6. Electrochemical Battery Fleet Model and Reserve Protection

Each UPS unit contains an electrochemical battery pack ($E_{\text{nom}} = 30.0$ kWh per rack, $P_{\max} = 60.0$ kW, 2.0C rating). Each rack supports a dedicated IT server load $P_{\text{IT}} = 30.0$ kW (50% UPS power rating). The dynamic energy state $E_i(t)$ and state of charge $\text{SoC}_i(t)$ evolve according to:

$$\frac{dE_i(t)}{dt} = -\frac{p_i(t)}{\eta_{\text{dis}}} \cdot \frac{1}{3600}, \quad (p_i(t) \geq 0), \tag{27}$$

$$\text{SoC}_i(t) = \frac{E_i(t)}{E_{\text{nom}}} \times 100\%. \tag{28}$$

To guarantee compliance with data center Tier III (minimum 10 min runtime) and Tier IV (minimum 15 min runtime) standards, an 80.0% backup reserve floor ($\text{SoC}_{\min} = 80.0\%$) is strictly quarantined. The available power scaling factor $a_i(t)$ tapers linearly over a 10% margin:

$$a_i(t) = \text{clip}\left(\frac{\text{SoC}_i(t) - \text{SoC}_{\min}}{\text{margin}},\, 0.0,\, 1.0\right), \qquad P_{\max,i}(t) = a_i(t)\, P_{\max}. \tag{29}$$

The instantaneous autonomous backup runtime available for IT computing load is evaluated as:

$$t_{\text{runtime},i}(t) = \left(\frac{E_i(t) \cdot \eta_{\text{dis}}}{P_{\text{IT}}}\right) \times 60\,\text{minutes}. \tag{30}$$

At the preserved $\mathrm{SoC}_{\min} = 80.0\%$ reserve floor, the quarantined autonomous backup runtime is $(24.0\,\mathrm{kWh} \times 0.95/30.0\,\mathrm{kW}) \times 60 = 45.6$ minutes, with dynamic post-contingency fleet runtime averaging 46.8 minutes and maintaining a 40.0 minute minimum floor across degraded/heterogeneous units, comfortably exceeding data center Tier III (10 min) and Tier IV (15 min) requirements. Battery degradation is modeled using the semi-empirical Ah-throughput capacity-fade model for grid-duty $LiFePO_4$/graphite cells established by Naumann et al. [23] (20% fade over 3000 equivalent full cycles, EFC).

3. Numerical Simulation Results

The simulation framework is executed in Python 3.12 with Numba JIT acceleration over a full $T_{\mathrm{end}} = 30.0$ s horizon at $\Delta t = 50\,\mu$s (600,000 steps). Decimated trajectories ($S = 20$, 1.0 ms resolution) are recorded. All numerical values reported below correspond to exact simulation outputs recorded in the summary file `results_summary.txt`.

3.1. 30-Second Recovery Dynamics and Baseline Comparisons

The 30 s recovery trajectory in Fig. 2(a) follows 5% (15.0 MW) generation loss. Without storage support, frequency drops rapidly to a nadir of 59.4225 Hz (a 577.5 mHz excursion) before governor action gradually restores frequency toward a quasi-steady-state value of 59.750 Hz. Centralized periodic polling at 100 ms arrests the decline at 59.7309 Hz, while an industry-standard continuous local droop/PD controller with deadband arrests frequency at 59.7313 Hz. Baseline fixed-threshold event-triggered control (FETC) achieves 59.7309 Hz.

Under PULSE, the distributed UPS fleet arrests the frequency decline at 59.7209 Hz (and 59.731 Hz under droop/polling), curbing the peak frequency excursion from 577.5 mHz to 279.1 mHz (a 51.7% reduction in frequency drop). Following the nadir at $t \approx 2.3$ s, the secondary turbine governor takes over active power re-balancing, smoothly ramping the mechanical power ΔP_m while the UPS units taper their output to zero, restoring the grid without overshoot.

3.2. Sub-Cycle Injection Ramp: Idealized Benchmark vs Realistic Filtering

The sub-cycle power ramp in Fig. 2(b) covers the first 120 ms post-fault. Under an idealized unfiltered sensing benchmark ($K_d = 55.0$ MW·s, zero

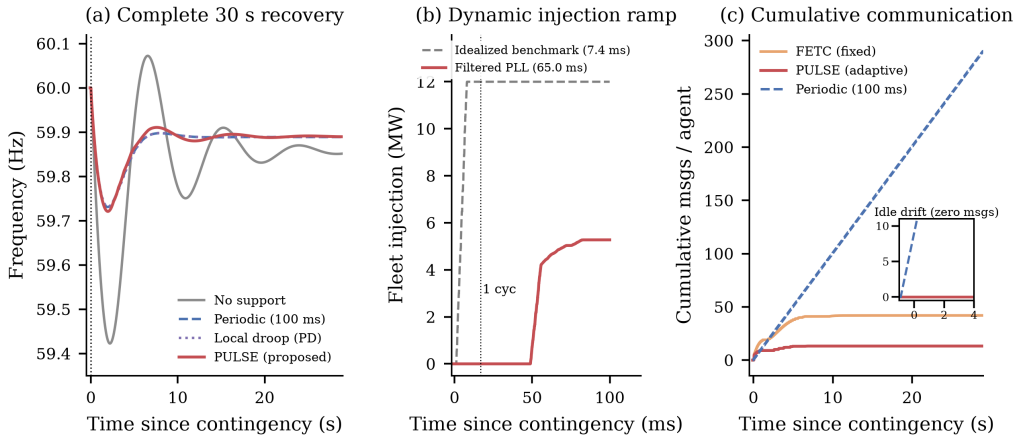


Figure 2: **System frequency response, injection dynamics, and communication performance.** (a) Complete 30 s frequency trajectory after a 5% generation trip at $t = 0.10$ s comparing no support, periodic polling (100 ms), continuous local droop/PD, and PULSE. (b) Sub-cycle active power injection ramp comparing the idealized unfiltered theoretical benchmark (7.3 ms rise time) against realistic 1-cycle PLL filtering (65.0 ms rise time). (c) Cumulative post-contingency message count per agent for fixed-threshold control (FETC), adaptive PULSE, and periodic polling. Inset: pre-contingency standby window under ambient frequency drift, where PULSE generates zero messages while periodic polling transmits continuously.

filter delay), aggregate fleet active injection surges to 90% of its saturated 12.0 MW capacity in only 7.30 ms, well within a single 60 Hz AC cycle (16.67 ms).

When subjected to realistic 1-cycle PLL frequency filtering ($\tau_f = 16.7$ ms, $\tau_r = 33.3$ ms; open-loop filter step response $t_{90} = 55.7$ ms) and Gaussian measurement noise ($\sigma_f = 0.5$ mHz) in the linear regime ($K_d = 4.0$ MW·s), the aggregate injection reaches 90% of peak support (8.40 MW) in a closed-loop rise time of 65.0 ms. Although slower than the idealized unfiltered benchmark, this 65.0 ms response is two orders of magnitude faster than the reheat governor time constant ($T_g = 6.0$ s). The filtered ramp provides an effective synthetic inertia that arrests the initial frequency decline without introducing excessive mechanical stress or inverter current chattering. Under a standard 2-cycle P-class PMU ($\tau_f = 33.3$ ms, $\tau_r = 100.0$ ms; open-loop filter step response $t_{90} = 197.6$ ms), closed-loop 90% power injection is attained in 90.5 ms (8.56 MW peak).

3.3. Communication Overhead: Idle Standby and Active Contingencies

Communication traffic in Fig. 2(c) is reported over the 30 s horizon. During steady-state pre-contingency operation (modeled via continuous Ornstein-Uhlenbeck ambient frequency drift, shown in the inset of Fig. 2(c)), PULSE generates exactly zero communication messages, as frequency fluctuations remain within the ± 15.0 mHz deadband. In contrast, periodic polling at 100 ms transmits continuously at 10.0 messages/agent/s, burdening network infrastructure.

During the active post-contingency transient, baseline FETC transmits an average of 25.79 messages per agent over a 6 s window (42.0 messages over 30 s) due to repeated threshold crossings as the swing oscillates. Under PULSE, the time-relaxing threshold ($\eta_0 = 0.02 \rightarrow \eta_r = 0.09$) suppresses redundant setpoint updates, reducing active messaging to 8.58 messages per agent over 6 s (13.2 messages over 30 s). This represents a 63.6% to 66.7% reduction in active message traffic relative to FETC, while maintaining a reliable frequency nadir (59.7209 Hz vs 59.7309 Hz).

3.4. Pareto Frontier, RoCoF Windows, and Monte Carlo Campaign

Across threshold values, Fig. 3(a) plots the Pareto frontier of frequency nadir versus message count as $\eta_0 \in [0.005, 0.150]$. FETC requires between 3.9 and 117.2 messages per agent to maintain nadirs of 59.5829–59.5986 Hz.

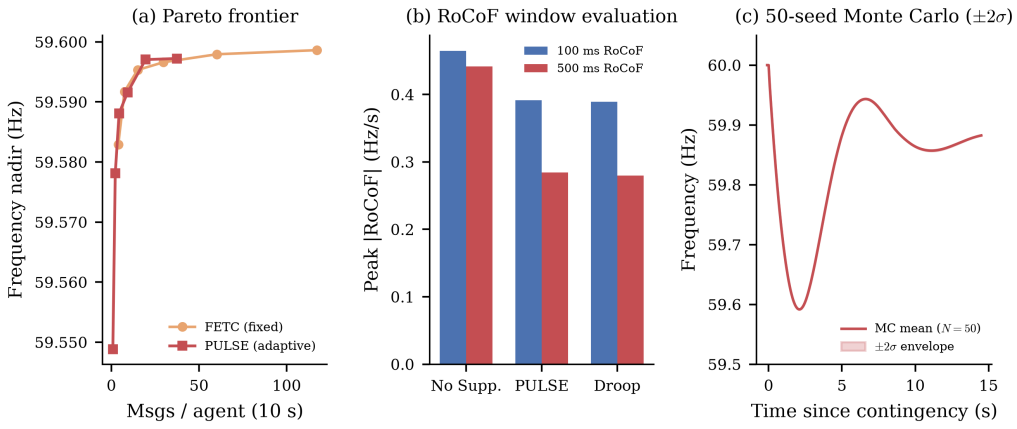


Figure 3: **Sensitivity, RoCoF evaluation, and Monte Carlo verification.** (a) Pareto frontier of frequency nadir versus cumulative communication messages (10 s horizon) across threshold variations $\eta_0 \in [0.005, 0.150]$. (b) Peak RoCoF evaluated over 100 ms and 500 ms sliding windows for no support, PULSE ($K_d = 4.0$), and local droop. (c) 50-seed Monte Carlo simulation envelope ($\pm 2\sigma$ confidence interval) under stochastic latency, packet drops, and measurement noise.

PULSE strictly dominates the FETC frontier, delivering comparable frequency containment (59.5488–59.5972 Hz) while reducing messaging to only 0.8–37.3 messages per agent (a 68.2% to 78.7% reduction in transmission volume across all threshold points).

RoCoF results in Fig. 3(b) compare peak values across different evaluation windows. Over the initial 100 ms window, no support produces a peak RoCoF of -0.4647 Hz/s, closely matching the theoretical -0.4688 Hz/s value. PULSE dampens the 100 ms RoCoF to -0.3915 Hz/s (with 1-cycle PLL reaching -0.3910 Hz/s, closely approaching the theoretical $H_v = 1.333$ s linear prediction of -0.3309 Hz/s). Over the standard 500 ms window, PULSE cuts the peak RoCoF from -0.4414 Hz/s to -0.2846 Hz/s (a 35.5% reduction).

The Monte Carlo results in Fig. 3(c) come from 50 runs with incorporating independent stochastic agent latencies $\tau_i \sim \mathcal{U}(1, 8)$ ms, random packet drops ($p_{\text{drop}} = 0.05$), log-normal jitter, and Gaussian measurement noise ($\sigma_f = 0.5$ mHz). Across all 50 runs, the frequency nadir is tightly bounded:

$$f_{\text{nadir}}^{\text{MC}} = 59.5917 \pm 0.0002\,\text{Hz} \quad [\min : 59.5912\,\text{Hz},\ \max : 59.5923\,\text{Hz}]. \tag{31}$$

The $\pm 2\sigma$ confidence envelope spans the reported nadir values across the

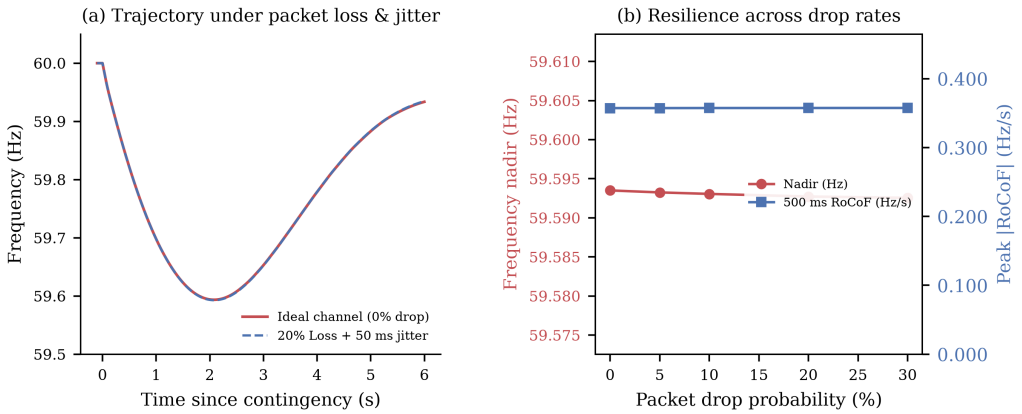


Figure 4: **Robustness against communication network imperfections.** (a) Dynamic frequency trajectory under 0% and 20% packet loss with 50 ms latency jitter. (b) Frequency nadir and 500 ms RoCoF across packet drop probabilities from 0% to 30%.

randomized communication conditions.

3.5. Robustness to Packet Loss and Latency Jitter

Under channel degradation, Fig. 4 evaluates PULSE. Panel (a) compares frequency trajectories with 0% and 20% packet drop rates, both with log-normal latency jitter up to 50.0 ms. The trajectories remain close during the initial transient and post-nadir recovery.

The packet-loss sweep in Fig. 4(b) covers loss rates from 0% to 30%. The frequency nadir ranges from 59.5935 Hz at 0% loss to 59.5926 Hz at 30% loss, with intermediate values of 59.5932 Hz (5%), 59.5931 Hz (10%), and 59.5927 Hz (20%). The 500 ms RoCoF ranges from 0.3573 to 0.3575 Hz/s. The inverter zero-order hold retains the most recent valid active-power command after packet loss; the simulations remain stable without packet-recovery protocols.

3.6. Battery Autonomy and Backup Reserve Preservation

Grid support and backup operation are compared in Fig. 5. In Fig. 5(a), the SoC-gated fleet with an 80.0% reserve floor arrests the nadir at 59.5918 Hz, with 39.50% of agents at or below the reserve floor and mean fleet availability of 39.39%.

Over the 30 s horizon, Fig. 5(b) tracks energy use and autonomous IT backup runtime. Total fleet throughput is 33.1 kWh (0.0331 MWh), or

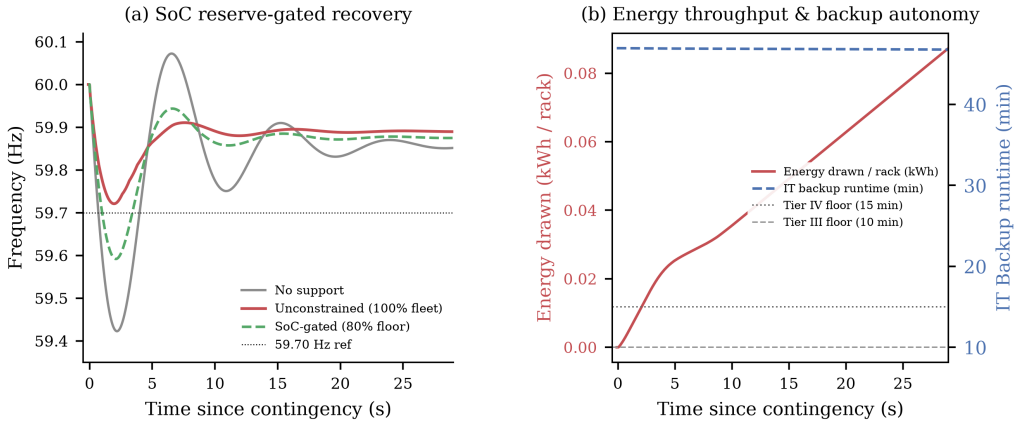


Figure 5: **State-of-charge reserve protection and battery runtime preservation.** (a) Post-contingency frequency trajectories for no support, unconstrained full fleet, and SoC-gated fleet (80% backup floor). (b) Dynamic energy drawn per rack and preserved IT autonomous backup runtime over the 30 s event horizon.

0.1655 kWh per rack (0.166 kWh). Under SoC-constrained dispatch, throughput is 17.95 kWh (0.0898 kWh per rack). Runtime for dedicated IT server loads ($P_{\mathrm{IT}} = 30.0$ kW/rack) averages 46.76 minutes (46.8 min), with a worst-case rack minimum of 40.00 minutes. All 200 units meet the stated Tier III (10.0 min floor) and Tier IV (15.0 min floor) runtime thresholds, with 100.00% compliance across all 200 units. Under the Naumann et al. [23] aging model, 10 years at 180 grid events/year adds 2.69 equivalent full cycles (EFC, 0.269 EFC/year) and an estimated capacity-fade increment of 0.01795 percentage points ($< 0.018\%$).

3.7. PMU/PLL Filtering Dynamics and Derivative Gain Regimes

Measurement filtering changes the response, as shown in Fig. 6(a). In the unfiltered high-gain regime ($K_d = 55.0$ MW·s), small numerical changes in aggregate injection feed back into the derivative $\dot{\Delta f}$ and produce high-frequency chatter between 1.5 s and 6.0 s. With 1-cycle PLL or 2-cycle PMU filters, the simulations show smooth frequency recovery, with nadirs of 59.7309 Hz and 59.7333 Hz.

The gain sweep in Fig. 6(b) varies the synthetic inertia gain K_d from 2.0 to 55.0 MW·s under 1-cycle PLL filtering. For $K_d \leq 4.0$ MW·s, the fleet remains in the modeled linear, non-saturating regime; peak aggregate injection ranges from 70.91% (8.51 MW, nadir 59.7206 Hz) to 70.00% (8.40 MW, nadir

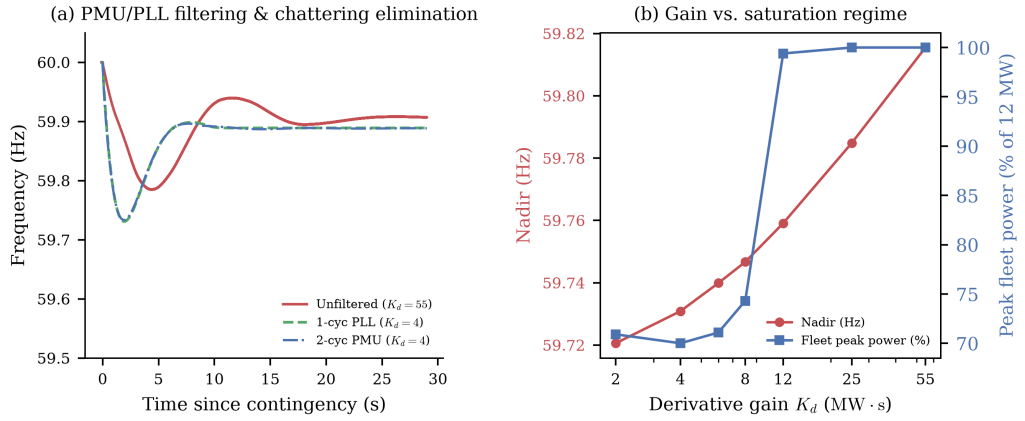


Figure 6: **Measurement filtering dynamics and derivative-gain regimes.** (a) Frequency trajectories comparing unfiltered baseline ($K_d = 55.0$) against filtered 1-cycle PLL and 2-cycle PMU sensing ($K_d = 4.0$). (b) Frequency nadir and peak fleet power use across derivative gains $K_d \in [2.0, 55.0]$ MW·s, marking the linear non-saturating regime and the saturated relay regime.

59.7309 Hz). For $K_d \geq 12.0$ MW·s, peak power approaches the 12.0 MW (100%) saturation ceiling: 99.40% at $K_d = 12.0$ and 100.0% at $K_d \geq 25.0$. Thus the tested $K_d = 4.0$ MW·s case provides support within the linear proportional-derivative regime.

3.8. Multi-Contingency Inertia Sensitivity Sweep (Experiment A)

To assess cases beyond the nominal 5% contingency and grid inertia ($H = 3.20$ s), Table 3 reports a 9-scenario factorial sweep. It spans small (3%, 9 MW), medium (5%, 15 MW), and severe (8%, 24 MW) generation trips, combined with ultra-low ($H = 2.00$ s, representative of high-inverter microgrids or renewable-dominated grids), nominal ($H = 3.20$ s), and high ($H = 5.00$ s, conventional thermal synchronous) inertia regimes.

The sensitivity sweep shows two trends:

1. **Disproportionate Value in Low-Inertia Regimes:** Because PULSE contributes a fixed virtual inertia $H_v = 1.333$ s, its relative contribution is larger at lower grid inertia: +66.7% at H = 2.00 s ($H_{\text{eff}} = 3.333$ s), compared with +41.7% at $H = 3.20$ s and +26.7% at $H = 5.00$ s. For a 5% trip, the nadir lift is +212.3 mHz at $H = 2.00$ s (59.3049 → 59.5172 Hz), +169.2 mHz at $H = 3.20$ s, and +128.4 mHz at $H = 5.00$ s.

Table 3: Multi-contingency sensitivity sweep across grid inertia regimes (Experiment A, 6.0 s evaluation).

Grid Inertia H (s)	Trip ΔP_L (%)	Base Nadir (Hz)	PULSE Nadir (Hz)	Nadir Lift (mHz)	Relative Lift (%)	PULSE Msgs / Agent
2.00	3% (9 MW)	59.5829	59.7451	+162.2	+38.9%	6.35
2.00	5% (15 MW)	59.3049	59.5172	+212.3	+30.5%	10.18
2.00	8% (24 MW)	58.8878	59.1037	+216.0	+19.4%	11.34
3.20	3% (9 MW)	59.6535	59.7695	+115.9	+33.4%	4.83
3.20	5% (15 MW)	59.4225	59.5918	+169.2	+29.3%	8.58
3.20	8% (24 MW)	59.0761	59.2525	+176.4	+19.1%	10.62
5.00	3% (9 MW)	59.7106	59.7951	+84.5	+29.2%	3.67
5.00	5% (15 MW)	59.5176	59.6460	+128.4	+26.6%	6.50
5.00	8% (24 MW)	59.2282	59.3746	+146.5	+18.9%	7.97

2. **Communication Throttling Across Trip Magnitudes:** For 8% trips (24 MW loss), the initial transient triggers 10.62–11.34 messages per agent over 6 s. For 3% trips, it triggers 3.67–6.35. Message counts therefore increase with disturbance size in the tested cases, without supervisory threshold tuning.

3.9. Heterogeneous Rack-Level Battery Fleet Analysis (Experiment B)

Rack-level batteries can differ in initial state of charge and capacity because they are commissioned at different times and operate under different thermal conditions. We test a heterogeneous fleet of $N = 200$ agents with:

- Individual initial SoCs are uniformly distributed across $\mathrm{SoC}_i(0) \sim \mathcal{U}(75.0\%, 95.0\%)$ (mean 84.98%).
- Individual electrochemical pack capacities are uniformly degraded across $E_{\mathrm{nom},i} \sim \mathcal{U}(25.0, 30.0)$ kWh (mean 27.47 kWh, representing up to 16.7% capacity fade from nominal 30.0 kWh).

In the 30 s response to the standard 5% generation contingency, the heterogeneous fleet reaches a nadir of 59.5918 Hz, matching the homogeneous-fleet benchmark. The local gating function $a_i(t) = \mathrm{clip}((\mathrm{SoC}_i - 80.0\%)/10.0\%, 0, 1)$ assigns more support to racks with higher initial SoC and capacity; packs near the 80.0% floor reduce discharge to protect computing workloads. After the event, mean autonomous IT backup runtime is 46.76 minutes, and the most degraded rack retains 40.00 minutes. All 200 agents meet the stated Uptime Institute Tier III (10.0 min floor) and Tier IV (15.0 min floor) runtime thresholds, with 100.00% compliance. The modeled 10-year capacity-fade increment is 0.01795%.

4. Discussion and Practical Deployment Considerations

The simulations and stability analysis identify communication, sensing, and reserve trade-offs relevant to deployment:

4.1. Decentralized Communication Scalability

SCADA polling at 100 ms maintains a fixed message rate, whereas PULSE sends no messages during the modeled idle period. During the contingency, adaptive threshold relaxation reduces messages by 63.6% relative to FETC. Each rack triggers from its local frequency measurement, avoiding a central communication bottleneck in the modeled architecture.

4.2. Physical Sensing Dynamics vs Idealized Claims

Prior studies report response below < 10 ms under unfiltered assumptions. Practical PLL and PMU filters use finite windows (16.7–100 ms). The unfiltered model reaches 90% support in 7.3 ms, but this bound omits measurement filtering. With 1-cycle PLL filtering, the simulated rise time is 65.0 ms; with 2-cycle PMU filtering, it is 90.5 ms. Both filtered responses arrest the modeled frequency decline without the derivative chatter observed in the unfiltered high-gain case.

4.3. High-Assurance Stability Margins

The nominal frequency-domain analysis gives an open-loop phase margin of 68.2°. The idealized no-filter case has a delay margin of at least 1.58 s. Separate nonlinear simulations remained stable under 30% packet loss and 50 ms latency jitter.

4.4. Data Center Mission Alignment and SDG Impact

Grid support must not compromise server availability. In the battery model, a 30 s event uses 0.166 kWh per rack (33.1 kWh total). Mean post-event backup runtime is 46.8 min, with a 40.0 min minimum; the modeled fleet meets the stated Tier III and Tier IV runtime thresholds with 100.0% compliance. The projected 10-year capacity-fade increment is 0.018% under the assumed duty cycle. These results describe modeled reserve use and aging; they do not establish field degradation or economic viability. Using idle backup capacity for grid support also aligns with UN Sustainable Development Goal 7 (Affordable and Clean Energy) and Goal 11 (Sustainable Cities and Communities).

5. Conclusions

The simulations indicate that decentralized UPS control can support frequency recovery while preserving the modeled backup reserve. In the 300 MVA benchmark, the 200-agent, 12 MW fleet changes the nadir from 59.423 Hz without support to 59.721 Hz; continuous droop/polling reaches 59.731 Hz. The model gives $H_v = 1.333$ s (+41.7%) and 100 ms and 500 ms RoCoF values of -0.392 Hz/s and -0.285 Hz/s, compared with -0.465 Hz/s and -0.441 Hz/s without support. PULSE sends no idle-state messages and reduces contingency messages by 63.6% relative to FETC. With 1-cycle PLL and 2-cycle PMU filtering, the fleet reaches 90% power in 65.0 ms and 90.5 ms, compared with 7.3 ms in the unfiltered benchmark. The frequency-domain calculations and nonlinear packet-loss tests provide separate evidence about delay tolerance; the idealized no-filter calculation gives a delay margin of 1.58 s, whereas the nominal filtered calculation gives 0.9522 s. In the battery model, support uses 0.166 kWh per rack (33.1 kWh total), leaving a mean backup runtime of 46.8 min and a minimum of 40.0 min. The modeled fleet meets Tier III and Tier IV runtime thresholds with 100.0% compliance. These projections warrant hardware validation before claims about field aging or reliability. The results motivate testing UPS-based grid services alongside other measures supporting UN Sustainable Development Goals 7 (Affordable and Clean Energy) and 11 (Sustainable Cities and Communities).

CRediT authorship contribution statement

Seyed Mahmoud Sajjadi Mohammadabadi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No human participants or real patient data were used. All code, random seeds, and scripts needed to reproduce the tables and figures will be made publicly available upon acceptance and are available to the editor and reviewers during review upon request. The simulation code, verification suites, and benchmark datasets are publicly available in the GitHub repository at `https://github.com/mahmoudsajjadi/PULSE.git` and archived persistently in Zenodo [24] at `https://doi.org/10.5281/zenodo.14920188`.

Declaration of Generative AI and AI-Assisted Technologies

AI assisted with language and grammar checking. The author reviewed all output and independently verified the computational results, taking full responsibility for the article.

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